Complete Study Roadmap: ECE → AI/ML Engineer (Data Analyst as Parallel Track)

For a 3rd-year ECE (Data Science specialization) student, moderate Python, rusty math

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HOW TO USE THIS DOCUMENT

Each phase lists: **what to learn → book(s) → free course → paid course (optional) → YouTube/video → project checkpoint.** Don't jump ahead. Don't collect resources without finishing one first. Pick ONE resource per row — you don't need all of them, they're alternatives.

Total timeline: ~14 months (aligned to placement season).

PHASE 0 — Math & Python Foundation (Weeks 1–6)

Python (proper, not just syntax)

- **Book:** "Python Crash Course" by Eric Matthes (beginner-friendly, practical)
- **Free course:** CS50P (Harvard's "Introduction to Programming with Python") — edx.org, free
- **YouTube:** freeCodeCamp's full Python course (single 4-hour video)
- **Checkpoint project:** Build a small CLI tool (e.g., expense tracker) using functions, file I/O, and error handling.

Linear Algebra

- **Book:** "Linear Algebra and Its Applications" by Gilbert Strang (the classic)
- **Video (intuition first):** 3Blue1Brown — "Essence of Linear Algebra" playlist (YouTube, free, ~2 hours total, watch this BEFORE the book)
- **NPTEL:** "Linear Algebra" by Prof. Bhaskar Bagchi / or MIT OpenCourseWare 18.06 (Gilbert Strang's own lectures, free on YouTube)

Probability & Statistics

- **Book:** "Introduction to Probability" by Blitzstein & Hwang (Harvard Stat 110 textbook — very readable)
- **YouTube (best resource, don't skip): StatQuest with Josh Starmer —** his stats fundamentals playlist is widely considered the clearest explanation available

anywhere, free.

- **NPTEL:** "Probability and Statistics" — IIT Kharagpur (search NPTEL portal, multiple offerings each semester)

PHASE 1 — SQL + Data Handling (Weeks 4–10, can overlap Phase 0)

- **Free course:** Mode Analytics SQL Tutorial (mode.com/sql-tutorial) — free, practical, industry-style
- **Practice platform:** LeetCode SQL section (free tier) + StrataScratch (real interview questions from FAANG/product companies)
- **Book:** "SQL for Data Analysis" by Cathy Tanimura
- **Pandas/NumPy:** Official pandas "10 minutes to pandas" doc, then Kaggle Learn's free "Pandas" micro-course (kaggle.com/learn)
- **Checkpoint project:** Take a messy public dataset (Kaggle has hundreds), clean it in pandas, write 10 SQL queries answering business questions about it.

PHASE 2 — Core Machine Learning (Weeks 8–18)

- **Course (the industry-standard starting point):** Andrew Ng's **Machine Learning Specialization** (DeepLearning.AI on Coursera) — paid (~₹2,000-3,000/month, financial aid available), but genuinely worth it for structure and rigor.
- **Free NPTEL alternative:** "Introduction to Machine Learning" by Prof. Balaraman Ravindran, IIT Madras — free to audit, ₹1000 for certificate exam (proctored, held periodically through the year). Also available: Prof. Sudeshna Sarkar's version from IIT Kharagpur, and a Tamil-medium version by Prof. Arun Rajkumar if that helps.
- **Book (the industry bible for hands-on ML):** "Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow" by Aurélien Géron — this single book will take you from theory to working code faster than anything else.
- **YouTube (Indian context, practical, free):** Krish Naik's ML playlist, or CampusX (both explain concepts with Indian job-market framing).
- **Practice:** Start Kaggle competitions NOW — Titanic, House Prices (Advanced Regression) — do them as you learn each algorithm, not after finishing the course.
- **Checkpoint project:** One complete end-to-end ML project — data cleaning → EDA → model → evaluation → write-up on GitHub with a README.

PHASE 3 — Deep Learning (Weeks 16–24)

- **Course:** DeepLearning.AI's **Deep Learning Specialization** (Coursera, paid) by Andrew Ng — covers neural nets, CNNs, RNNs.
- **Free alternative (more code-first):** fast.ai's **"Practical Deep Learning for Coders"** — completely free, taught top-down (build first, theory after), excellent if you learn better by doing.
- **NPTEL:** "Deep Learning" — IIT Madras/Kharagpur offerings (free to audit; ₹1000 certificate exam).
- **Book:** "Deep Learning" by Goodfellow, Bengio, Courville (the reference textbook — dense, use for concepts you need to go deeper on, not as your primary learning path).
- **Checkpoint project:** An image classifier or a text classifier using PyTorch or TensorFlow, deployed as a simple demo.

PHASE 4 — GenAI / LLMs / Agents (Weeks 22–32) — YOUR DIFFERENTIATOR

This is the part almost nobody in your batch will have done properly. Prioritize it.

- **Free course:** Hugging Face's **NLP Course** (huggingface.co/learn) — free, hands-on, covers transformers properly.
- **Free short courses (1-2 hrs each, all free):** DeepLearning.AI's short course library — specifically "LangChain for LLM Application Development," "Building and Evaluating Advanced RAG," and "AI Agents in LangGraph."
- **Paid (if budget allows):** DeepLearning.AI's full Generative AI specializations, or Scaler's AI Engineering course (IIT Roorkee collaboration) if you want a structured, mentored program with placement support.
- **Tools to actually get your hands dirty with:** OpenAI API or Anthropic API (both have free/cheap tiers for learning), LangChain/LangGraph, Chroma (free, local vector database — best for learning before you touch Pinecone/Qdrant).
- **YouTube:** Andrej Karpathy's "Neural Networks: Zero to Hero" and his "Let's build GPT" video — genuinely the best deep-dive on how LLMs work internally, free.
- **Checkpoint project (the big one):** Build a RAG application — e.g., "chat with your college syllabus/notes" — with a working demo link. This single project matters more than any certificate on this list.

PHASE 5 — Deployment, Git, MLOps Basics (Weeks 30–38)

- **Git/GitHub:** "Git and GitHub for Beginners" — freeCodeCamp (YouTube, free). You should already be using this since Phase 1 — this is just formalizing it.
- **Docker:** Docker's own "Getting Started" guide (free, docs.docker.com) — you only need basics: build an image, run a container.
- **Deployment:** Streamlit (docs.streamlit.io) to turn any Python model into a clickable web app in an afternoon — do this for every project going forward.
- **Cloud (pick one, don't do all):** AWS Free Tier + freeCodeCamp's "AWS Certified Cloud Practitioner" video course, OR Google Cloud's free Qwiklabs intro labs.
- **Checkpoint project:** Take one earlier project and fully deploy it — GitHub repo + Docker + live demo link (Streamlit Cloud/Render are free hosting options).

PHASE 6 — Data Analyst Track (Parallel, Weeks 6–16) — for early internships

- **Power BI:** Microsoft's free "Power BI for Beginners" learning path on Microsoft Learn (learn.microsoft.com) — official, free, well-structured.
- **Tableau:** Tableau's free public training videos (Tableau Public + their "Tableau for free" YouTube series).
- **Book:** "Storytelling with Data" by Cole Nussbaumer Knaflic — how to make visuals people actually understand, widely used in analyst interview prep.
- **Checkpoint project:** One polished dashboard (Power BI or Tableau) on a real dataset with a written business insight summary — this alone can get you a data analyst internship.

PHASE 7 — Interview Prep & Portfolio (Weeks 38–52, ongoing)

- **DSA basics (not competitive programming — just enough for interviews):** "Striver's SDE Sheet" (takeuforward.org, free) — focused, not overwhelming.
- **ML interview prep:** "Ace the Data Science Interview" by Nick Singh & Kevin Huo (book) — covers SQL, stats, ML, and behavioral questions asked at real companies.
- **Mock interviews:** Pramp (free peer-to-peer mock interviews) or Interviewing.io.
- **Portfolio:** GitHub with 3-4 projects (one classic ML, one deep learning, one RAG/LLM app, one deployed end-to-end) + a simple portfolio site (GitHub Pages, free) linking to all of them.
- **LinkedIn:** Post about your projects as you build them — recruiters in this space actively source from LinkedIn posts showing real project work, not just profiles.

QUICK REFERENCE: Free vs Paid Priority

If budget is tight, these are 100% free and sufficient to get you job-ready: NPTEL (ML + DL), fast.ai, Hugging Face NLP course, Kaggle Learn, StatQuest, 3Blue1Brown, Krish Naik/CampusX, Mode SQL tutorial, freeCodeCamp, DeepLearning.AI's free short courses.

Worth paying for if you can (better structure, mentorship, projects): Andrew Ng's ML + DL Specializations (Coursera, ~₹2-3k/month), Scaler's AI Engineering course (higher cost, but includes mentorship + placement support — evaluate based on your budget and whether you need structured accountability).

THE ONE RULE THAT MATTERS MOST

Every phase above has a "checkpoint project" — do not skip these to move faster through courses. Recruiters and interviewers care far more about 3-4 real, documented, deployed projects than about how many courses you've completed. If you only have time for one thing this year, prioritize finishing Phase 4's RAG project — it's the single highest-leverage item on this entire list.